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DSA Survey Strategy and Coverage Goals

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1. CASS Survey Design and Implementation

The Deep Synoptic Array (DSA) will be a groundbreaking new radio survey telescope anticipated to begin operations in 2029. It will operate in the 0.7 – 2 GHz band with parallel observing modes and unprecedented survey speed to deliver science-ready polarimetric, spectrally resolved image cubes, pulsar timing data, and pulsar/FRB discovery data.

A key component of the DSA observing plan is a cadenced all-sky survey (CASS), covering the entire sky north of declination δ≥ -30° (30,940 square degrees), with a variable depth and cadence strategy. Fundamental questions surrounding the baryon cycle in galaxies, the formation of stars over cosmic time, the influence of active supermassive black holes (SMBHs) on galaxies and the nature of dark matter and dark energy, will be addressed by detecting over a billion star-forming galaxies and active SMBHs, and by observing the neutral-hydrogen kinematics and contents of several million galaxies. The array will revolutionize the field of radio transients, detecting >100,000 fast radio bursts (FRBs), >20,000 pulsars and >1 million slow transients, with sub-arcsecond localization for host galaxy identification. The CASS strategy was iteratively developed together with a survey implementation plan designed to maximize survey uniformity, sensitivity, and operational efficiency. We will summarize the key outcomes of that optimization.

1 1.1. Tiling and Mosaicking Strategy

The tiling and mosaicking strategy is outlined in detail in the survey implementation plan, with key outcomes summarized here. The DSA Radio Camera will produce calibrated images for individual pointings, each of which has a standard duration of 21 minutes, determined to be the optimal observation duration to deliver the necessary uv-coverage for the radio camera, with a resulting output data rate of ~10 GB/s. The baseline image size is 17,456 x 17,456 pixels with 0.8” pixels at 1.35 GHz, ensuring coverage down to the ~10% power point of the primary beam. Image size is fixed across frequency, with the pixel size and primary beam varying inversely with frequency. Images are combined using a weighted linear mosaic to create the final sky maps at each frequency. The wide bandwidth (0.7–2.0 GHz) presents challenges due to the varying antenna primary beam width, with the area of the primary beam varying by a factor of 8.16 across the full frequency range of the telescope. To achieve uniform sensitivity across the sky and the frequency band, a hexagonal packing pattern for the pointings has been adopted. The design utilizes a Modified Hexagonal Pattern with a pointing separation of θhex=1.667° and a row separation of θrow=1.45°. This optimized overlap results in an average weight factor (Zmean) of 1.44 at 1.35 GHz. The effective integration time in the final continuum mosaic is therefore 44% longer than the individual pointing duration. At the bottom of the band (0.7 GHz) there is substantial overlap of each pointing, while at the top of the band the hexagonal separation (e.g. in RA) is equal to the primary beam width at this upper frequency. Uniformity at all frequencies can be further improved in the final co-added survey data through variation of the pointing map between epochs.

To manage observations and data processing efficiently, the ~12,800 pointings required to cover the full sky area are grouped into a "Tiered-Tile" scheme. The adopted design uses 5x6

Tiles (~30 pointings per tile). This results in 427 unique tiles covering the survey area, with each tile representing approximately 72.5 deg². Assuming a 60% time allocation, and assuming 80% observing efficiency to account for slewing between pointings, down-time for maintenance and poor observing conditions, the total budget over five years is 2135 tile visits allocated across the 427 unique tiles, or an average of 5 visits per tile. However, CASS will employ a variable depth and cadence strategy, prioritizing specific higher priority footprints for more observing time and prioritizing observations in the declination range where the telescope has a higher survey speed.

2. Allocation of Tiles to Survey Area

2 2.1. Scientific Motivation for Variable Depth and Variable Cadence

The key science with the DSA, together with the performance of the telescope, requires several considerations in survey design that favor varying the prioritization of observing time across the sky.

Fundamentally, there is the goal of providing a sensitive radio survey covering the entire available sky. This will include both the extragalactic sky and Galactic plane, but as discussed below, the cadence requirements are somewhat different for imaging of the Galactic plane. As well as providing an essential broad reference for the entire astronomical community, this directly supports several Key Science Goals (KSGs), e.g., mapping the Galactic 21-cm hydrogen line emission in the Galaxy and its halo (KSG3), providing a deep reference map for the detection of the counterparts to compact binary coalescences detected in gravitational waves (KSG2), and tracing star formation, neutral gas reservoirs and cosmic ray production in local universe galaxies (KSG3, KSG4), and for identifying dark low-mass dark matter halos that retain pristine HI but lack any stars, known as Reionization-Limited H I Clouds (RELHICs) (KSG7).

Simultaneously, much of the key science with the DSA relies on the association with an optical/infrared counterpart and in many cases requires good redshift determination when applied in a statistical sense to larger samples. This will be important for identifying host galaxies to fast radio bursts (KSG5), stellar explosions (KSG6), and the counterparts to compact binary coalescences detected in gravitational waves (KSG2). It will also be essential for identifying the multi-wavelength counterparts to strong lensing candidates (KSG7), for characterizing the occurrence of AGN and star formation activity throughout cosmic time (KSG4) and for identifying dark low-mass dark matter halos that retain pristine HI but lack any stars, known as Reionization-Limited H I Clouds (RELHICs).

Finally, the telescope's sensitivity, and therefore survey speed, varies as a function of zenith angle due to spillover noise contribution to the telescope system temperature (T_{sys}). The sensitivity varies by 38% over the zenith angle range, and therefore declination range, used in the survey. Survey speed is proportional to sensitivity squared and therefore varies by a factor of 1.9 over the declination range to be used in the survey. Generally speaking, allocating more time to declinations where the telescope has better survey speed maximizes the scientific return

of the telescope. Note that increased confusion noise and less efficient multi-wavelength cross matching capability due to coarser spatial resolution will further degrade sensitivity at large zenith angles for continuum data products.

Quite aside from the sensitivity requirements for the survey, the multi-epoch nature of the survey, and associated cadence requirements, are fundamentally driven by time domain applications across a range of key science cases. This includes including time domain cosmography of AGN (KSG7), the discovery of lensed fast radio bursts (FRBs; KSG5), discovery and characterization of pulsars (KSG8) and discovery and characterization of variables and transients varying on timescales ranging from minutes to years (KSG6). A uniform cadence within each observing tile would be poorly suited to maximizing the scientific return from the survey. A complex and varied observing cadence designed to be sensitive to phenomena across a wide range of timescales is required. The detailed planning of the cadence for DSA observations is an ongoing research project for a number of graduate students and involves simulation of each population to determine the spacing of observations required to maximize overall science return.

2.2. The Euclid and HSC-Wide Footprint (~10,000 deg^2)

CASS prioritizes overlap with the Euclid survey area in the northern hemisphere (~9,000 deg^2) and the Hyper-Suprime-Cam Wide Survey (HSC; ~1400 deg^2) for deeper observations with the DSA.

The specific footprints chosen are:

- HSC-Wide: The North, Spring Equatorial, and Fall Equatorial fields (Aihara et al. 2018) defined by:
  - North: 13h20m≤RA≤16h40m, +42∘≤Dec≤+44.5∘
  - Spring: 09h00m≤RA≤15h30m, −2∘≤Dec≤+5∘
  - Fall: 22h00m≤RA≤02h40m, −1∘≤Dec≤+7∘
- Euclid North: The Northern 'mainland' and 'island' (Scaramella et al. 2022). This region is defined by Ecliptic latitude $|\beta| > 10°$ and avoiding low Galactic latitude ($|b| \leq 23°$, or $|b| \leq 25°$ if Dec ≥ +30°).

Euclid will provide spectroscopic redshifts for ~35 million galaxies (~20 million in the DSA footprint) using the Near-Infrared Spectrometer and Photometer (NISP) instrument in its slitless spectroscopy mode. These galaxies will primarily lie in the redshift range between z=0.84 and z=1.88, reflecting the primary goal for the spectroscopic survey of measuring baryon acoustic oscillations (BAO) and redshift-space distortions (RSD). Simultaneously, Euclid will provide photo-z for the ~1.5 billion galaxies that constitute its weak lensing sample down to a magnitude of 24.5. The target precision for individual photometric redshifts is $\sigma_z/(1+z)<0.05$ for galaxies out to a redshift of z≈2. However, the precision will vary as a function of redshift. The 4000 Å Balmer break is redshifted into the wavelength region between Euclid's VIS and Y-band filters leading to poor redshift determination in the redshift range z≈1.5−2.5 (the so-called redshift desert). Euclid will therefore require the addition of multi-band optical data from the ground (*u, g,*

r, i, z bands), particularly from Rubin, to break the degeneracies inherent to Euclid's filter set alone.

The Hyper-Suprime-Cam Wide Survey provides five-band optical coverage (*g, r, i, z, y*) with photo-z calibrated and validated against a wealth of deep spectroscopic data from other surveys (e.g., zCOSMOS, DEEP2, VIPERS, GAMA, SDSS). The redshift precision for bright galaxies in their sample (i<25) is typically ~5%.

The combined ~10,000 deg^2 footprint of Euclid and HSC-Wide will have 7 epochs of DSA observations, with a predicted thermal noise for the combined map <400 nJy.

2.3. The Galactic Plane and Bulge (~5,500 deg^2)

The Galactic plane represents a second area of heightened priority for additional epoch allocation in the CASS. This is defined as Galactic latitude |b|<10∘, expanded to |b|<15∘ in the Galactic Bulge region (|l|<20∘). The primary goal is to identify and confirm Galactic pulsars, with a secondary goal of identification of various classes of transients, stellar systems, and diffuse Galactic structures in the CASS image data products.

The DSA pulsar survey addresses key science surrounding tests of gravity, measurements of the equation of state of bulk nuclear matter, and pulsar timing array searches for gravitational waves (KSGs 8 and 1). A wide range of ancillary science programs are supported, such as a comprehensive measurement of the neutron-star velocity distribution, the determination of the magnetic-field and ionized-gas structure of the Galaxy, a measurement of the occurrence rate of binary neutron star systems in different environments, and a census of the endpoints of massive-star evolution.

The DSA pulsar survey will accomplish this by performing by far the most sensitive search for pulsars yet conducted and is forecast to increase the known population by a factor of seven. The survey will run commensally with every pointing. However, the increased epoch allocation in the Galactic plane and bulge is designed to enable the survey to target and characterize >90% of the predicted canonical *and* millisecond pulsar population; the latter population motivates the bulge extension.

The Chronoscope backend will be used to conduct two independent pulsar searches simultaneously, commensally with the Radio Camera imaging.

In a targeted search, a list of candidate pulsars will be defined using an initial CASS epoch to create a deep continuum point source catalog. From this catalog, pulsar candidates will be selected by first removing extended radio sources and then cross-matching with OIR galaxy catalogs to filter out extragalactic objects such as AGNs and star-forming galaxies. For each pointing, the system will form up to 4000 synthesized beams targeting these candidates and analyze the data across approximately 500 Dispersion Measure (DM) trials and 500 acceleration trials to identify periodic signals, including from binary systems. In a blind search, 2e5 beams will be formed that tile part of the field of view, which will be directly searched for periodic / accelerated signals over 60s. During a 1260s pointing, the entire field of view can be searched by progressively adjusting which beams are searched.

3 The targeted search is anticipated to be highly efficient in regions of the sky where diffuse Galactic emission is not exceedingly bright. However, in the inner few tens of deg in Galactic longitude and for Galactic latitudes |b|<~3 deg, the blind search will be required to reach the expected sensitivity threshold. This is the region with the highest predicted pulsar density on the sky.

Pulsar candidates require follow up observations for both confirmation and characterization. Thanks in particular to the enhanced number of epochs on the Galactic plane and bulge, this

will be done commensally as well. The survey will leverage the CASS's densely sampled, hexagonal pointing grid. Pulsar candidates can be followed up either in subsequent epochs or, more efficiently, in adjacent pointings within the same observing block. An individual observing block may occupy a few weeks. The strategy can be optimized by placing the tunable 312.5 MHz search band at lower frequencies (around 700-1100 MHz), where the larger primary beam creates more overlap between pointings. This allows a candidate to be re-observed multiple times in quick succession, maximizing on-source time and increasing the cumulative sensitivity by a factor of up to 7 for some beams. A subset of the 4000 available search beams per pointing will be specifically allocated to re-observe candidates in a coherent fold mode found in the outer parts of the primary beam from a previous pointing. These data will be used to confirm pulsar discoveries, and to conduct (low-precision) timing observations to determine pulsar spin periods, period derivatives, binarity, and variability. Discoveries of high scientific interest will be studied in more detail via the discretionary time allocation. The Galactic plane represents another area of heightened priority for additional epoch allocation in the CASS. This is defined as Galactic latitude $|b|<10^\circ$, expanded to $|b|<15^\circ$ in the Galactic Bulge region ($|l|<20^\circ$). The primary goal is to identify and confirm Galactic pulsars, with a secondary goal of identification of various classes of transients in the CASS image data products.

2.4. Wide Area Tiers (~15,500 deg^2)

Outside the high-priority footprints, CASS time is allocated in a fashion that delivers the overall sky area defined by the survey (30,940 deg^2) while prioritizing those areas where survey speed is highest.

Epochs are allocated in the remainder of the sky outside the high-priority footprints in three tiers, defined by fixed declination bands. This strategy provides an operationally simple way to invest more observing time where the telescope has the highest survey speed:

- **4 Epochs (Low Zenith Angle)**: Covers the declination range of +17.4° to +60.9°.
- **3 Epochs (Medium Zenith Angle)**: Covers the intermediate declination ranges.
- **2 Epochs (High Zenith Angle)**: Covers the most extreme declination ranges of < -8.7° and > +69.6°.

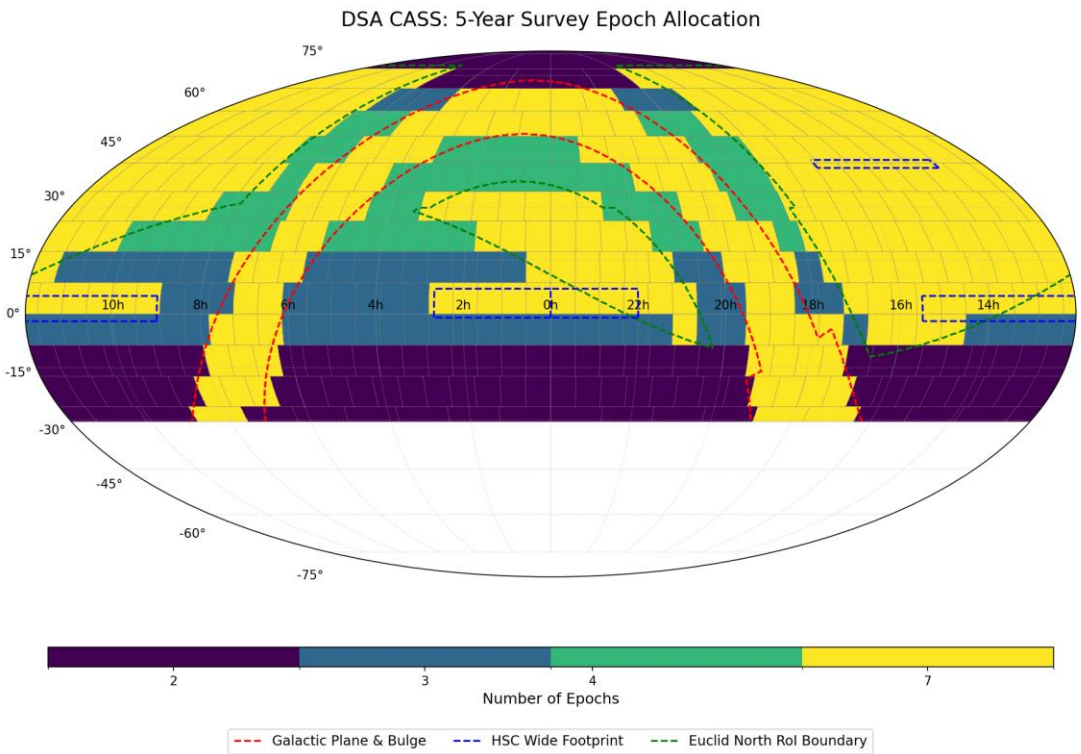

Figure 1. The number of epochs allocated within the 30,940 deg^2 survey area of the Cadenced All Sky Survey (CASS). The overlapping footprint with the Euclid survey and the HSC-Wide survey, as well as the Galactic plane and Bulge, are designated high priority. For the remaining ~50% of the sky, epochs are allocated in fixed declination bands, with more observing time allocated where the telescope has the highest survey speed.

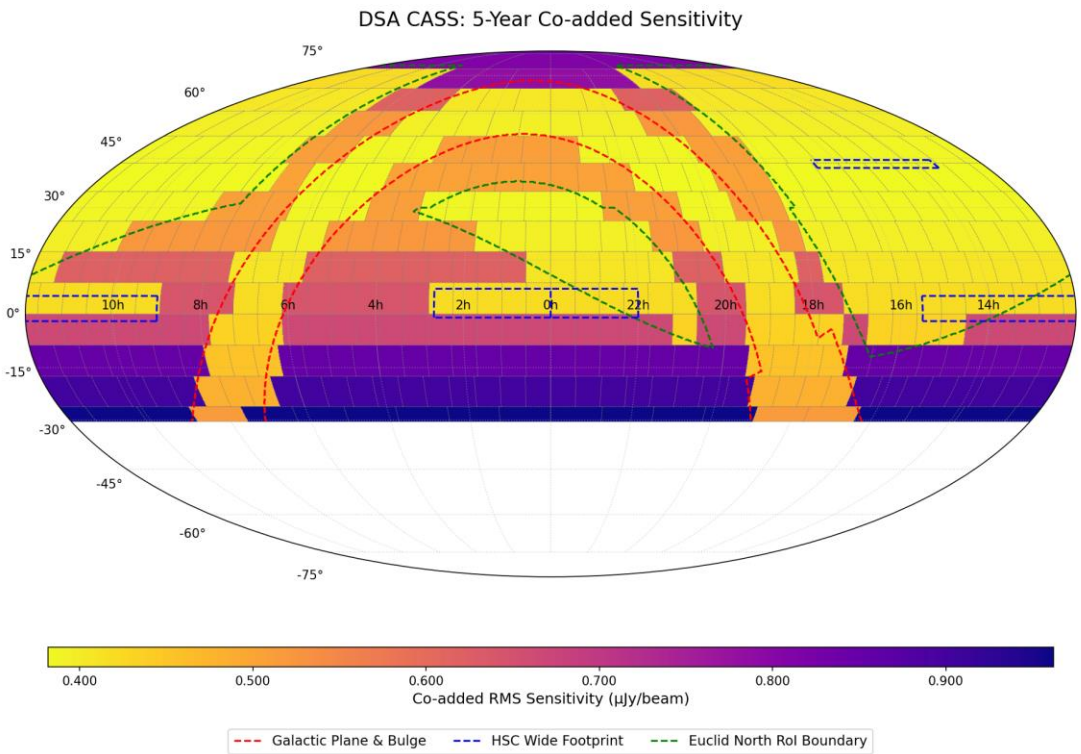

Figure 2. The final sensitivity of the co-added CASS survey. This conservatively assumes a T_{sys}/ε at declination $\delta = 0$ of 25K/0.7, where ε is aperture efficiency, an 80% observing efficiency and that 65% of the band is not occupied by RFI. This model does not yet account for increased contribution to T_{sys} from Galactic synchrotron at low Galactic latitude and does not fully capture the impact of confusion noise at large zenith angles.

3. Implications for Data Rate and Volume

We briefly highlight the implications of the proposed survey design for the data rate out of the radio camera, as well as the final data volume to be stored long-term. For all observations, the Radio Camera will produce 12,774 images consisting of a range of continuum, spectral, and polarimetric data products. Simultaneously, the Chronoscope will search the full field (lower ~25% of the band) for fast time domain transients such as fast radio bursts and will perform both the targeted and blind pulsar searches described above. The output from the radio camera dominates the overall data volume and data rate to the post-processing (DAT) subsystem. Each image is 17,456 x 17,456 pixels, with the pixel size inversely proportional to frequency, varying from 1.55” to 0.54” from the bottom to the top of the band. For each observation, the 12,774 images are a combination of continuum, spectral line and polarimetric products:

Wideband Continuum (WC) images - These cover the full 0.7-2.0 GHz band (Stokes I/V), using 10 x 130.2 MHz channels. There will also be Full-bandwidth Continuum (FC) products combining these channels into single images in Stokes I flux density and spectral index.

Medium-wide Polarimetric (MP) images - using 500 x 2.604 MHz in Stokes IQUV. A single co-added representation of the sky would be accumulated for each of the MP products, or possibly the set of Tiled mosaic images.
Spectral Line: Narrow Line (NL) and Zoom A & B (ZA, ZB) images - HI spectroscopic data at three different spectral (velocity) resolutions.

- NL: 5600 channels at 130.2kHz resolution tracing HI out to *z* ~1 (0.7-1.43GHz).
- ZA: 4192 channels at 8.138kHz covering nearby extragalactic HI (1388 - 1422 MHz).
- ZB: 960 channels at 1.017kHz covering Galactic HI (1419.93-1420.91 MHz).

Each 17,456 x 17,456 pixel image is ~1.2 GB in size. Each pointing produces 12774 images and there are 12,800 pointings required to cover the full survey area. Therefore, the raw image data for this full survey, with an average of 5 epochs per tile, is 1 EB. The continuum data are a small fraction of this volume and will therefore be archived per pointing and can be made available to a public archive in both single pointing and mosaiced format. The spectral line and polarimetric data dominate the volume and are reduced to a manageable size in two steps. The combination of data into mosaics and finally tiles leads to a substantial reduction, and the spectral line and polarimetric data are not stored per epoch, but rather as a single co-added representation of the sky that is accumulated in each epoch. This reduces the data volume (pre-compression) to a manageable 50 PB. These full accumulated maps are not sent to the public archive. Rather image cubes (resolved galaxies) and spectra (unresolved sources) are extracted based on *a priori* source lists (e.g., known galaxies/sources in multi-wavelength surveys) and sources identified in DSA images. These extracted data are <1 PB in size and will be sent to the Schmidt Sciences public archive.

The data rate from the radio camera to post-processing (DAT), and the size of compute for post-processing (mosaicking, deconvolution, etc.) is similarly determined by the typical duration of each observation, and therefore the number of epochs. For a 21-minute duration observation, the data rate is 18 GB/s. The current chosen number of epochs for the DSA is chosen to balance the scientific needs of the project with the cost of data throughput to post-processing and the cost of compute for post-processing. A proposed design change to the radio camera is under consideration, whereby the uv grids, rather than image data, are sent to post-processing. The subsequent padding of the grid to full size and FFT to the image domain would then take place during the mosaicking step. This would significantly reduce the data rate to post-processing, opening the possibility of increasing the number of epochs if ongoing cadence studies present a strong motivation to do so.

Finally, we note that as much as 40% of the observing time will be allocated to observations of single fields, including targeted (e.g. pulsar timing) and Discretionary Time fields. For such observations, the radio camera will collect all continuum, spectral and polarimetric imaging products as well as fast time domain products. However, to ensure these data don't present a significant overhead in long term storage, only the extracted spectral and polarimetric products will be stored, unless the data are integrated into existing mosaics.